

Project Review Presentation on Maximization of Savings in Large Scale Radial Distribution Systems Using Hybrid PSO-SSA Algorithm

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Maximization of Savings in Large Scale Radial Distribution Systems Using Hybrid PSO-SSA Algorithm

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- ❖ Power system is the combination generation, transmission and distribution. Distribution system is the tail end which is supplying the electrical power to the consumers. The demand of electricity is rising day-to-day.
- ❖ Increasing of load in day to day will increase the impact on the distribution system.
- ❖ The thermal constraints of the existing distribution lines will increase the losses further.
- ❖ The modifications of the topology of the distribution system with the existing lines without including the new components in the system is called network reconfiguration.
- ❖ The no of tie line switches will depend upon the type of the system and loops existing in the system.

Introduction



- Distribution system reconfiguration is defined as altering the topological structure of distribution feeders by changing the open/closed states of the sectionalizing switches (normally closed) and tie switches (normally open).

Benefits from reconfiguration during normal operation :

- 1) Minimizing the system losses.
- 2) Load balancing among feeders.
- 3) Reliability indices improvement.
- 4) Increase loadability/DG penetration.

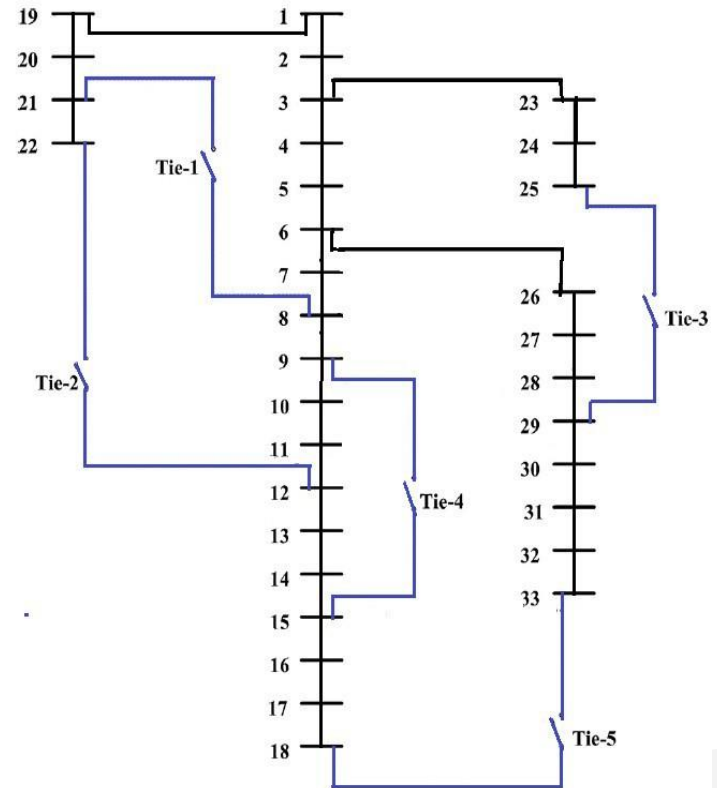


Fig-1 33 RDS

Literature Review

Sl. No	Authors & Year	Research Focus	Methodology Used	Key Contribution	Limitations
1	Salau et al., (2020)	Distribution network reconfiguration	Optimization-based network reconfiguration	Reduced power losses and improved voltage profile using reconfiguration strategy	Did not consider economic savings or cost-benefit analysis
2	Karunarathne et al., (2020)	Optimal DG sizing and placement	Comprehensive Learning Particle Swarm Optimization (CLPSO)	Improved voltage profile and reduced network losses with optimal DG placement	Focus limited to technical loss reduction, economic savings not maximized
3	Dhal et al., (2022)	Distribution network reorganization	Whale Optimization Algorithm (WOA)	Significant reduction in real power loss through network reconfiguration	Lack of economic optimization and hybrid optimization approach

Literature Review

Sl. No	Authors & Year	Research Focus	Methodology Used	Key Contributions	Limitations
4	Kumar and Rao (2018)	Network reconfiguration and DG placement	Analytical optimization method	Improved voltage profile and reduced losses	Did not consider integrated capacitor placement or savings maximization
5	Shaheen et al., (2022)	Network reconfiguration with DG and capacitor allocation	Artificial Ecosystem Optimizer (AEO)	Enhanced voltage stability and reduced losses using DG and capacitors	Focus mainly on technical optimization, limited economic savings modeling
6	Mahdavi et al., (2022)	Distribution system reconfiguration considering load pattern	Advanced consumption modeling and optimization	Improved accuracy of load modeling and network reconfiguration performance	No hybrid optimization approach and limited economic benefit analysis



Literature Review



The following points highlight some of the significant findings from the previous study:

- Existing research on distribution system reconfiguration and distributed generation placement primarily emphasizes technical objectives such as loss minimization and voltage profile enhancement.
- However, limited attention has been given to maximizing economic savings, including operational cost reduction and financial benefits. Moreover, most studies focus on single optimization techniques, which may not provide globally optimal solutions.
- The research gap in the literature motivates the study for contributing on modeling of RDS with reconfiguration, finding the novel methodology on formulating the hybridization with latest optimization techniques PSO & SSA, analyzing the large distribution networks which resembles the practical distribution networks.
- Therefore, there is a need to develop a hybrid optimization-based framework that integrates network reconfiguration, DG placement with the objective of maximizing economic savings and improving voltage profile and loss minimization.

Objectives of the Project

The objective of the research work is,

- To minimize the system losses in the Radial Distribution System.
- To improve the voltage profile of radial distribution system.
- To optimize the radial distribution system with the optimal reconfiguration and optimal location of DGs to maximize the loss cost savings by reducing the losses.
- Validation of the suitability of the proposed approach by implementing it on 119 and 135 Bus standard systems.

Load flow

- The line currents are calculated at backward stream using Kirchhoff's current law (KCL) and the node voltages are calculated at forward stream using Kirchhoff's Voltage law (KVL).
- The equations that are used for calculating power losses , branch currents and node voltages are shown below.
- $S_n = P_n + j Q_n = V_n \cdot I_{L,n}^*$ is the apparent power injected to bus 'n'
- The load current at any node is

$$I_n = \frac{P_{L,n} - j Q_{L,n}}{V_n^*}$$

- Where, $P_{L,n}$, $Q_{L,n}$ & V_n are the active power demand , reactive power demand , voltage magnitude at load bus respectively.
- The correlation among load currents & line currents by backward sweep as

$$[I_B] = [BIBC][I_L]$$



Load flow



Let V_a , V_b are the sending and receiving end of branch 'ab' with impedance Z_{ab}

The receiving end voltage by forward sweep as

$$V_b(k) = V_a(k) - I_b(k) Z_{ab}(k)$$

The power losses are calculated by

$$P_L = \sum_{b=1}^n I_b^2 Z_{ab}$$

Problem Formulation



The Optimization technique helps Maximize the Savings by reducing the losses

$$f_{objps0} = \min P_{real} * k_l (\$ / kw)$$

$$f_{objDA} = \min P_{real} * k_l (\$ / kwh) + \sum_{i=1}^{Ndg} DG_i * (k_{dg} (\$ / kw) + k_m (\$ / kwh))$$

Where

P_{real} active power losses in kW.

- k_l is loss cost coefficient 0.06 \$/kWh.
- k_{dg} is Installation cost coefficient 400 \$/kW
- k_m is Maintenance cost coefficient 0.045 \$/kWh

Proposed Methodology

HYBRID OPTIMIZATION TECHNIQUE (PSO-SSA):

The hybridization of algorithms is done as PSO-SSA. Among two algorithms PSO is used for placement for reconfiguration where as SSA is used for the placement and size of DG.

Proposed Methodology



The method for PSO-SSA is detailed as below.

Step.1: The maximum number of iterations, beginning and ultimate velocities, number of particles, number of searching agents, peri of SSA, and other PSO and SSA parameters should be initialized.

Step.2: Initialize the particles with branch numbers and searching agents with bus numbers and size of the DG.

Step3: Determine the system's losses by analysing the required radial distribution system's line and bus data.

Step.4: Consider the initial losses are optimum losses at zeroth iteration

$$P_{\min} = P_{\text{loss}}(0)$$

Step.5: Increment the population and searching agents

Proposed Methodology



Step.6: Determine the losses of the system with new combination of the switches and DG in the system.

Step-7: Compare the losses $P_{Loss}(1)$ with $P_{Loss}(0)$.

If $P_{loss}(0) > P_{loss}(1)$

$P_{min} = P_{loss}(1)$

Else

$P_{min} = P_{loss}(0)$

Step:7 Revise the PSO velocities with position values as shown in the equation.

$$U(\text{Revise}) = M * X + V_i * \text{rand}(1) * (P_{tidy} - U) + V_f * \text{rand}(1) * (G_{tidy} - U) \quad (5.10)$$

Step:8. Similarly update the position of the SSA algorithm parameters.

Step: 9 Repeat the step 2 to step 7 using load flows and determine the losses.

Step: 10 Display the result which are optimum reconfiguration and location of DG.

PSO		SSA	
Number of particles	30	Number of Searching Agents	40
Number of iterations	50	Number of iterations	50
Max inertia	0.9	-	-
Min inertia	0.2	-	-
Initial velocity	2		-
Final velocity	2	-	-

Table: 1 Algorithm parameters and their values

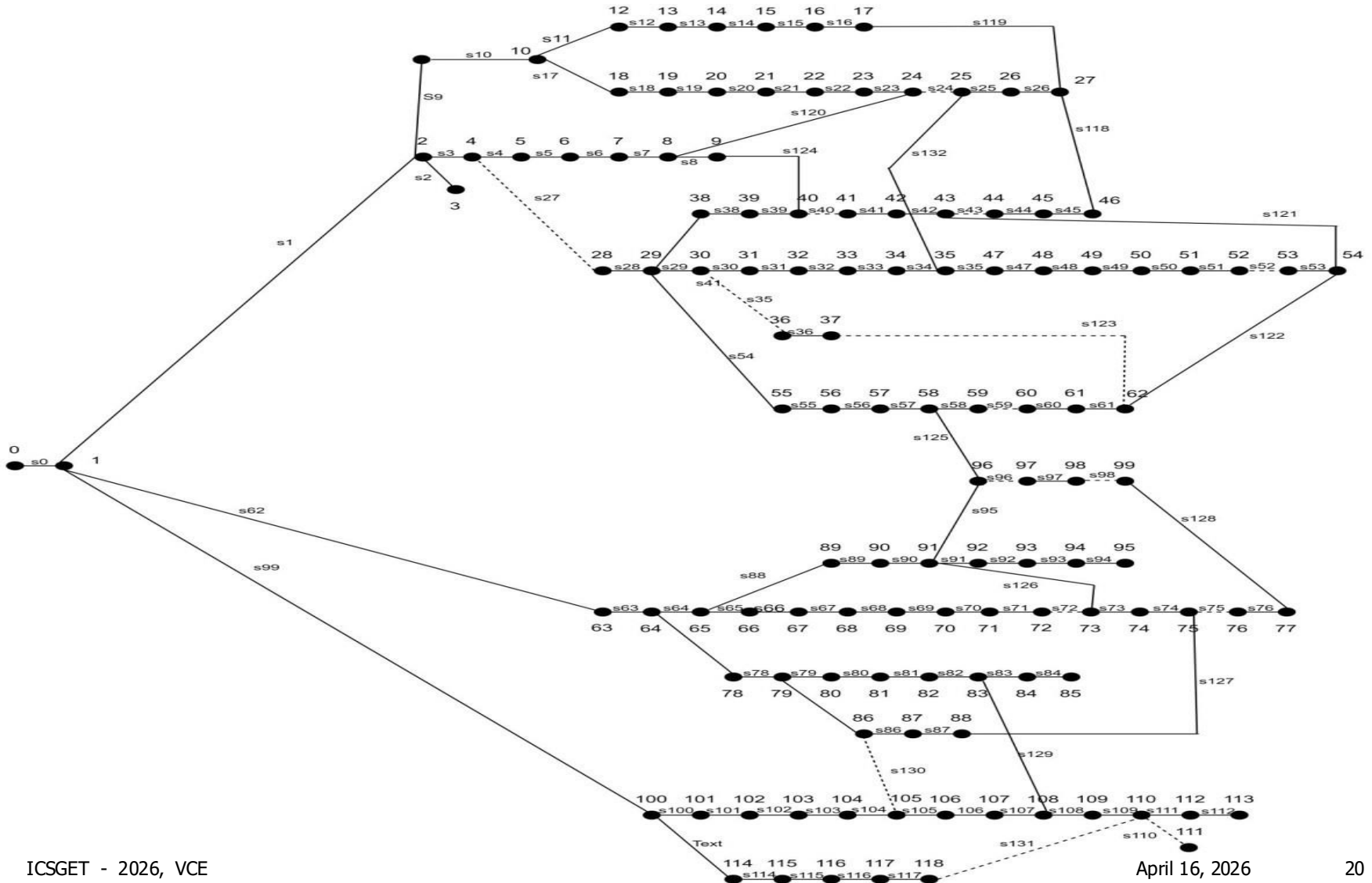
Results and Discussion

- The performance of the hybrid method PSO-SSA is tested on two test systems which are 119 bus and 135 RDS.
- Figure 2 shows the hybrid method's single-line reconfiguration schematic for the 119 bus. The true power loss, voltage profile, and loss cost for the 119-node RDS without and with DG location and PSO-SSA rearrangement are shown in Table 1.
- This approach optimizes the rearrangement using the PSO methodology and the DGs' location using SSA.



Results and Discussion

Test System-1: 119 bus system



Results and Discussion

S. No.		Base Case	Reconfiguration (without DG)	Reconfiguration (with DG)
1	Open switches	119-133	43,27,23,53,123,126,75,72, 129,130,131,132,33	43,27,23,53,123,126,75,7 2,129,130,131,132,33
2	Power loss (KW)	1298.01	885	633
3	% Loss reduction	-	31.82	51.23
4	Minimum voltage (p.u)(Bus No)	0.88(115)	0.931(115)	0.9478(113)
5	DG Placement (Bus No)	-	-	118
		-	-	99
6	DG Size (KW)	-	-	518(kW)
		-	-	1808(kVar)
7	Loss Cost (USD)	68,26,182	46,54,215	55,79,300

Table 2 Comparison of 119-Bus RDS Configurations: With and without DG (PSO-SSA)

- The true power loss, voltage profile, and loss cost for the 135-node RDS without and with DG location and PSO-SSA rearrangement are shown in Table 3. This approach optimises the rearrangement using the PSO methodology and the DGs' location using SSA.
- The input for the very beneficial location of DG utilizing SSA is the redesigned distribution system. The 99th bus, with a load of 1808 KVar, and the 118th bus, with a load of 518 kW, are the most beneficial DG spots, with power loss of 633 kW and a loss cost of USD 55,79,300. The cost to put up is 400 USD/kW. Finally, in terms of reorganization loss cost, the structure's economy is 10,74,915 USD/annum.

Power Loss(KW)

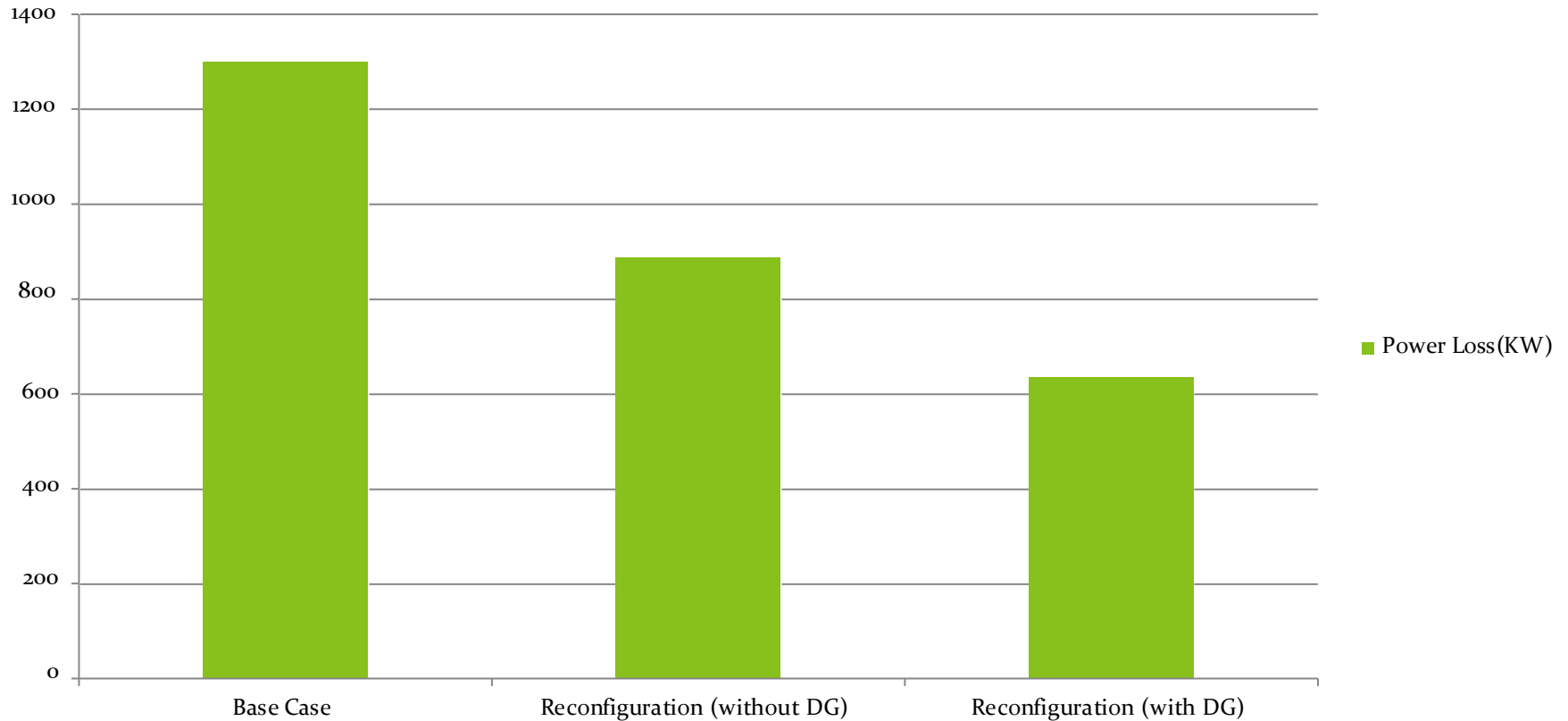


Figure 3 119-node system comparison
of Power loss for all cases

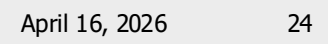


Figure 4 135-bus system reconfiguration with hybrid PSO-SSA method

S. No.		Base Case	Reconfiguration (without DG)	Reconfiguration (with DG)
1	Open switches	136 -156	7 35 51 90 96 106 118 126 135 137 138 141 142 144 145 146 147 148 150 151 155	7 35 51 90 96 106 118 126 135 137 138 141 142 144 145 146 147 148 150 151 155
2	Power loss (KW)	306	281	262
3	% Loss reduction	-	08.16	14.37
4	Minimum voltage (p. u)(Bus No)	0.938(117)	0.963 (106)	0.971(136)
5	DG Placement (Bus No)	-	-	104
		-	-	34
6	DG Size (KW)	-	-	58.6 (KW)
		-	-	908 (KVar)
7	Loss Cost (USD)	16,09,254	14,77,779	16,02,000

Table 3 135 bus RDS without and with DG placement along with reconfiguration (PSO-SSA)

- The true power loss, voltage profile, and loss cost for the 135-node RDS without and with DG location and PSO-SSA rearrangement are shown in Table 3. This approach optimises the rearrangement using the PSO methodology and the DGs' location using SSA.
- When all tie-line switches are open and sectionalized, the network loses 306 kW of power and costs 17,09,254 USD, or 0.06 USD/kWh. The network had a loss of 281 kW and a cost of 66,54,215 USD following the restructuring of 7, 35, 51, 90, 96, 106, 118, 135, 137, 138, 141, 142, 144, 145, 146, 147, 148, 150, 151, and 155.
- The input for the very beneficial location of DG utilizing SSA is the redesigned distribution network. With a loss decrease of 262 kW and a loss cost of USD 16,02,000. The most advantageous DG positions are on the 104th and 34th buses, with 58.6 kW and 908 KVar, respectively. The cost to put up is 400 USD/kW. Finally, when it comes to the cost of reorganization, the structure's economy is 75,779 USD per annum.

Y power Loss(KW)

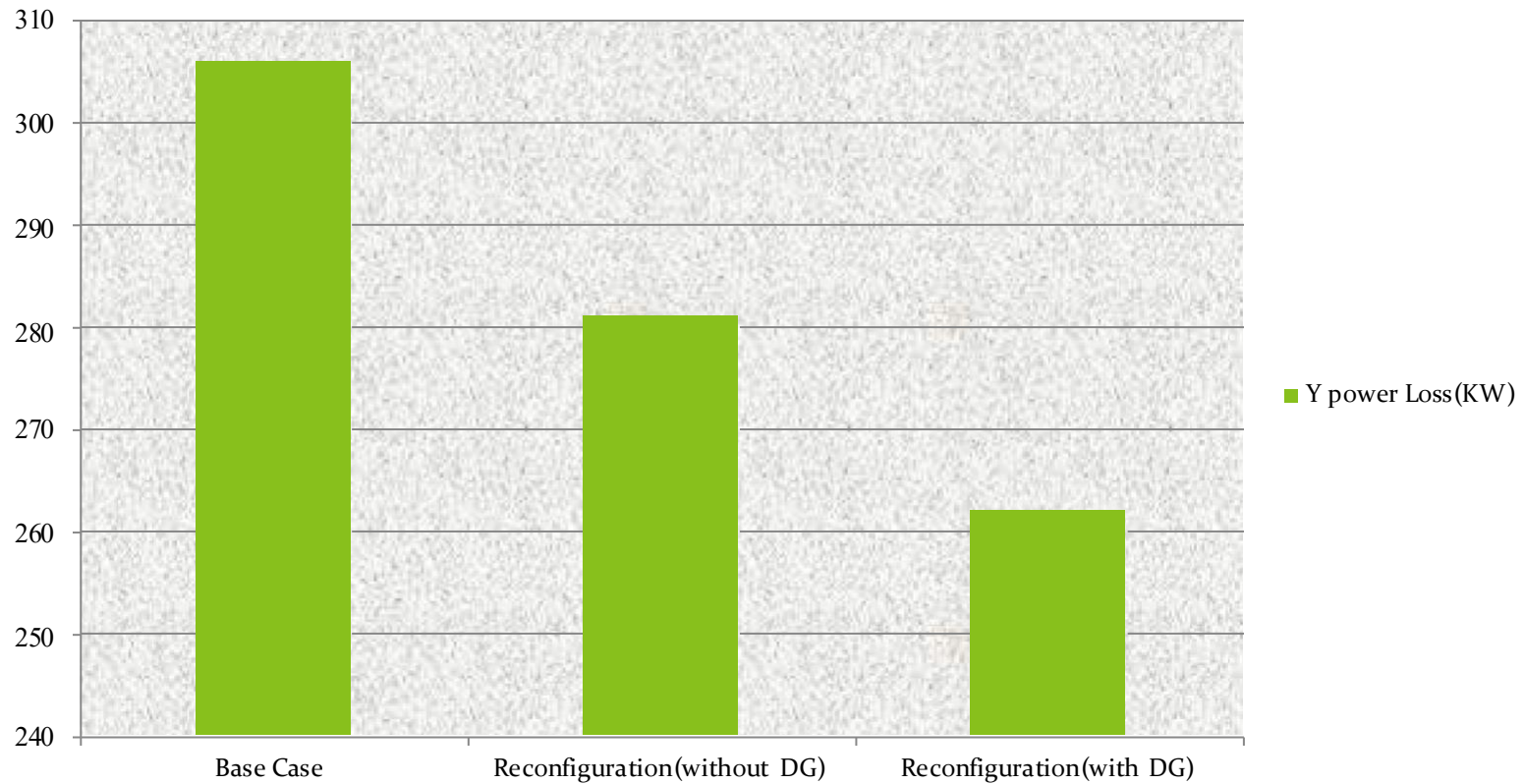


Figure 5 135-node system comparison of Power loss for all cases



Conclusions



- The hybrid optimization technique PSO-SSA is implemented for the optimal reconfiguration and optimal location of DGs to maximize the loss cost savings by reducing the losses.
- This method is tested on 119 and 135-node distribution networks.
- It is found PSO-SSA method real power losses are reduced from 1298.01 KW to 633KW i.e. reduced to 51.23%, Total Cost Savings of 10,74,915 USD /annum, whereas voltage profile is improved from 0.88 p. u to 0.9478 p. u for 119-bus System.
- For 135 bus system, real power losses are reduced from 306 KW to 262KW i.e. reduced to 14.37%, Total Cost Savings of 75,779 USD /annum, whereas voltage profile is improved 0.938 p. u to 0.971 p. u.



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CERTIFICATE OF PARTICIPATION

This is to certify that Dr./Mr./Ms. Rafi Vempalle, Hussain Vali Shaik, Harsha Vardhan Sugavasi, Anji Naik Bhukya, Likhitha Goddumarri & Arifa Begum Shaik of JNTUA College of Engineering, Pulivendula, India have presented a paper entitled Maximatization Of Savings In Large Scale Radial Distribution Systems Using Hybrid PSO-SSA Algorithm in 4th International Conference on Smart Grid and Green Energy Technologies during 13th and 14th March 2026 organized by Department of Electrical and Electronics Engineering, Vardhaman College of Engineering, Hyderabad.

Co-Ordinator

General Chair



Future Scope



The future scope of the work is discussed as follows,

- The proposed method is extended with demand side management by adopting smart technology solutions.
- The demand-side management is used at peak load and light load conditions with PV system includes at peak hours of the load.
- The method is further proposed for unbalanced systems.

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